Nanyang Technological University

EE6222 Machine Vision

Assignment 2: Donald Trump Fake Video Generation

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1 Data Preparation

1.1 Face Images

To generate realistic fake videos of Trump, high-resolution images of his faces with different perspectives are needed. I may get high quality dataset from open-source database such as authoritative image libraries and news agencies or mainstream media and documentaries. Images include frontal, profile, and tilted views with diverse lighting conditions.

Preprocessing Steps:

- Resize all images to 128×128 pixels (Unification),
- Normalize pixel values to fit model distribution (Normalization),
- Data augmentation: flipping, color jitter, and rotation to increase variability.

1.2 Motion Data (Video Clips)

Videos of Trump's exhibiting different movements (walking, talking, turning) and facial expressions in different conditions are extracted. Optical flow or keypoint-based motion representations are computed to provide temporal guidance.

2 Network Architecture

2.1 Overview

I proposed model is a **diffusion framework** for fake video generation based on **transformer**, which means the model will be trained to generate *Trump's video* by learning to denoise a sequence of temporally coherent frames sampled from Gaussian noise.
At each diffusion step, the network predicts the noise component to progressively reconstruct the clean video sequence. This architecture could emphasizes both **spatial appearance consistency** and **temporal motion coherence** [1] [2].

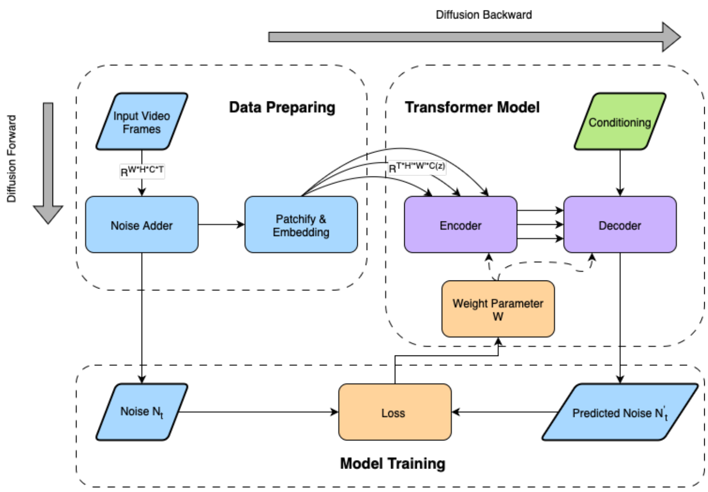

Figure 1: Frame of Video Generation Model

2.2 Input and Conditioning

Each training iteration samples a diffusion timestep t and constructs a noisy video sequence X_t as:

$$X_t = \sqrt{\alpha_t} X_0 + \sqrt{1 - \alpha_t}\,\epsilon, \quad \epsilon \sim \mathcal{N}(0, I),$$

where X_0 denotes the original clean video and ϵ is the added Gaussian noise. The model predicts the noise $\hat{\epsilon}_\theta(X_t, t)$ given:

- **Input:** a noisy video sequence $X_t = \{x_t^1, x_t^2, \ldots, x_t^T\}$ consisting of T frames,
- **Temporal embedding:** sinusoidal or learned positional encoding that encodes the timestep t,
- **Objective:** estimate the added noise and progressively reconstruct the clean sequence.

2.3 Diffusion Backbone with Transformer Denoiser

The denoiser follows a **U-shaped Transformer** backbone, combining both spatial-temporal self-attention and hierarchical feature aggregation.

- **Encoder:** extracts local and global representations from X_t. Each frame is first divided into non-overlapping patches, linearly projected into embeddings, and augmented with spatial and temporal positional encodings. The encoder applies alternating layers of *spatial attention* (within each frame) and *temporal attention* (across frames) to capture motion patterns and context.
- **Condition Injection:** conditioning information (class label or motion vector) is fused via **cross-attention** layers. The motion embeddings serve as queries to guide temporal consistency, while the class embedding modulates feature channels through **adaptive layer normalization (AdaLN)** or conditional scaling.
- **Decoder:** mirrors the encoder with skip connections to preserve fine-grained spatial details. The decoder outputs the predicted noise $\hat{\epsilon}_\theta$ of the same dimensionality as X_t, which is used for the reverse diffusion step.

The overall denoising process can be formulated as:

$$\hat{\epsilon}_\theta = f_\theta(X_t, t), \qquad X_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(X_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}}\,\hat{\epsilon}_\theta \right) + \sigma_t z,$$

where f_θ is the Transformer-based denoiser and $z \sim \mathcal{N}(0, I)$ is random noise.

2.4 Loss Functions

The training objective combines reconstruction and perceptual constraints:

- **Denoising loss:** the standard diffusion training objective encourages the model to predict the noise added to the video.
- **Perceptual loss:** computed in feature space (e.g., VGG or CLIP), ensuring texture fidelity and visual realism.
- **Temporal consistency loss:** penalizes frame-to-frame discrepancies to encourage smooth and coherent motion.

The total loss is a weighted sum:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{MSE}} + \lambda_1 \mathcal{L}_{\text{perceptual}} + \lambda_2 \mathcal{L}_{\text{temp}}.$$

3 Potential Improvements

Up till now, a deep learning-based fake video generation model has been established, which satisfies the basic requirements of the assignment. However, this model cannot generate personalized or controllable videos, meaning that the actions of Donald Trump in the generated video cannot be explicitly directed. To address this limitation and further enhance motion realism, a **joint motion conditioning module** can be introduced.

3.1 Motion Embedding from Text and Reference Video

Instead of relying solely on optical flow or 2D pose keypoints, the motion embedding can be extracted from:

- A **text description** of the desired action (e.g., "walking forward and raising right hand"),
- A **reference video** demonstrating the motion trajectory.

A lightweight Motion Encoder g_ψ converts these multi-modal inputs into a compact motion embedding:

$$m = g_\psi(\text{text description, reference video}).$$

3.2 Integration into Diffusion Backbone

The motion embedding m can be incorporated into the Transformer denoiser through mechanisms such as **Cross-Attention layers**, allowing the network to selectively attend to motion cues across time frames,

3.3 Improvements

Introducing motion conditioning via text and reference video provides several advantages:

- Enables more realistic and temporally coherent motion trajectories (e.g., walking, gesturing, turning),
- Preserves spatial identity and appearance consistency of Donald Trump across frames,
- Allows controllable video synthesis by modifying the text description or reference video, without retraining the entire diffusion model.

4 Resources & Infrastructure

4.1 Pre-trained Models & Tools

- **Diffusion Backbone:** pre-trained video diffusion models
- **Motion Encoder:** lightweight model to extract motion embeddings from optical flow, 2D/3D pose keypoints, reference video, or text description.
- **Evaluation Tools:** Pre-trained identity models (ArcFace) and optical flow / pose extraction libraries (e.g., RAFT, OpenPose, MediaPipe) for quantitative metrics.

4.2 Hardware & Data Pipeline

- High-performance GPUs (e.g., NVIDIA A100, RTX 4090) recommended for training high-resolution video sequences.
- Multi-GPU setups and mixed-precision training (FP16) to accelerate iterative denoising steps in diffusion models.
- Efficient data pipeline: PyTorch DataLoader with video augmentation (temporal cropping, horizontal flipping, color jitter) for large-scale batches.
- Adequate storage and memory for high-resolution videos and motion embeddings.

4.3 Evaluation Metrics

Video quality: FVD (Fréchet Video Distance), PSNR, SSIM, LPIPS to evaluate realism and fidelity.

Identity preservation: cosine similarity between generated and reference embeddings using pre-trained recognition models (e.g., ArcFace) to ensure appearance consistency.

Temporal consistency: Average Pose Distance (APD), frame-to-frame motion correlation, or optical flow alignment to quantify smooth motion.

5 Design Philosophy & Thoughts

5.1 Philosophy

- **Diffusion Backbone:** Leveraging iterative denoising in latent space ensures high-fidelity video frames. Temporal attention within the Transformer maintains smooth motion across frames.
- **Joint Motion Conditioning:** By embedding motion information from optical flow, 2D/3D pose sequences, reference videos, or textual descriptions, the network can generate temporally coherent and realistic motion trajectories. Cross-attention layers allow video features to selectively attend to motion cues, providing fine-grained control over generated actions.
- **Training Strategy:** Stage-wise training allows pre-training the diffusion backbone on large-scale video datasets, followed by fine-tuning with motion-conditioned inputs to achieve controllable video synthesis with minimal additional effort [3].

5.2 Conclusion

Recently, diffusion method has been a popular approach for image generation, which solves the collapse of classic GAN models. In addition, the trnsformer architecture could associate long-range dependencies. Therefore, I think the combination of diffusion and transformer may suitable to generate related high quality images, which is the frames of the video. This is the main reason why I select transformer archetecture to establish the diffusion backbone of video generator.

References

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[3] X. Peng, Z. Zheng, C. Shen, T. Young, X. Guo, B. Wang, H. Xu, H. Liu, M. Jiang, W. Li, Y. Wang, A. Ye, G. Ren, Q. Ma, W. Liang, X. Lian, X. Wu, Y. Zhong, Z. Li, C. Gong, G. Lei, L. Cheng, L. Zhang, M. Li, R. Zhang, S. Hu, S. Huang, X. Wang, Y. Zhao, Y. Wang, Z. Wei, and Y. You, "Open-sora 2.0: Training a commercial-level video generation model in $200k," 2025.